

EESC 214 Earth Science Data Analysis

Autumn Weaver

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1 Looking at Data

Objectives when first looking at data:

1. Understand the general character of the dataset
2. Understand the general behavior of individual parameters
3. Detect obvious problems with the data

Several steps when analyzing data:

1. Format conversion
Converting format to something easier to analyze. Format conversion is often needed but can introduce error.
2. Reality Checks Check that the data makes sense in context and compared to your expectations.
3. Time Plots Plot the data against time to observe trends.
4. Histograms Plot the distribution of data to observe patterns and potential errors or outliers.
5. Rate Information Plot the rate of change of different parameters.
6. Scatter Plots Plot parameters against each other to observe if there are any correlations.